



Southeast University

Introduction to Software Engineering

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Outline

Part 0. Overview

Part 1. Software Processes

Part 2. Modeling

Part 3. Quality Assurance

❖ **Chap 15-16. Software Quality**

❖ **Chap 17-19. Testing Strategy & Techniques**

❖ **Chap 20-21. Security Engineering & SCM**

Part 4. Software Project Management

Why Security Engineering?

- 保护系统资产免受攻击
- 修bug难以提升安全性

- ❖ Security is a prerequisite (前提) to system integrity, availability, reliability, and safety.
- ❖ **Security provides the mechanism that enable a systems to protect its assets from attack.**
- ❖ Assets are system resources (information, files, programs, storage, processor capacity) that have value to its stakeholders.
- ❖ Attacks take advantage of vulnerabilities (弱点) that allow unauthorized system access.
- ❖ It is difficult to make a system more secure by responding to bug reports, **security must be designed in from the beginning.**

- 安全性是系统完整性、可用性、可靠性和安全性的前提。
- 安全性提供了使系统能够保护其资产免受攻击的机制。
- 资产是对其利益相关者有价值的系统资源（信息、文件、程序、存储、处理器容量）
- 攻击利用允许未经授权系统访问的漏洞（弱点）。
- 通过响应错误报告来使系统更加安全是困难的，安全性必须从一开始就设计进来

Online Security Threats

- ❖ ***Social Media*** – networks often allow their users to develop applications that have access to personal details of their users
- ❖ ***Mobile Applications*** – native apps running on mobile devices may have the same access resources as the device owner
- ❖ ***Cloud Computing*** – brings additional confidentiality and trust issues into the security picture because it blurs the line between “trusted inside” and “untrusted outside”
- ❖ ***Internet of Things*** – ability of everyday objects to communicate and report contextual information about its user and its environment

- **社交媒体** - 网络通常允许用户开发能够访问其用户个人信息的应用程序。
- **移动应用** - 运行在移动设备上的本地应用可能拥有与设备所有者相同的访问资源。
- **云计算** - 带来了额外的机密性和信任问题，因为它模糊了“可信内部”和“不可信外部”之间的界限。
- **物联网** - 日常物品能够传递和报告关于其用户及其环境的上下文信息。

SOFTWARE CONFIGURATION MANAGEMENT

软件配置管理

The “First Law”

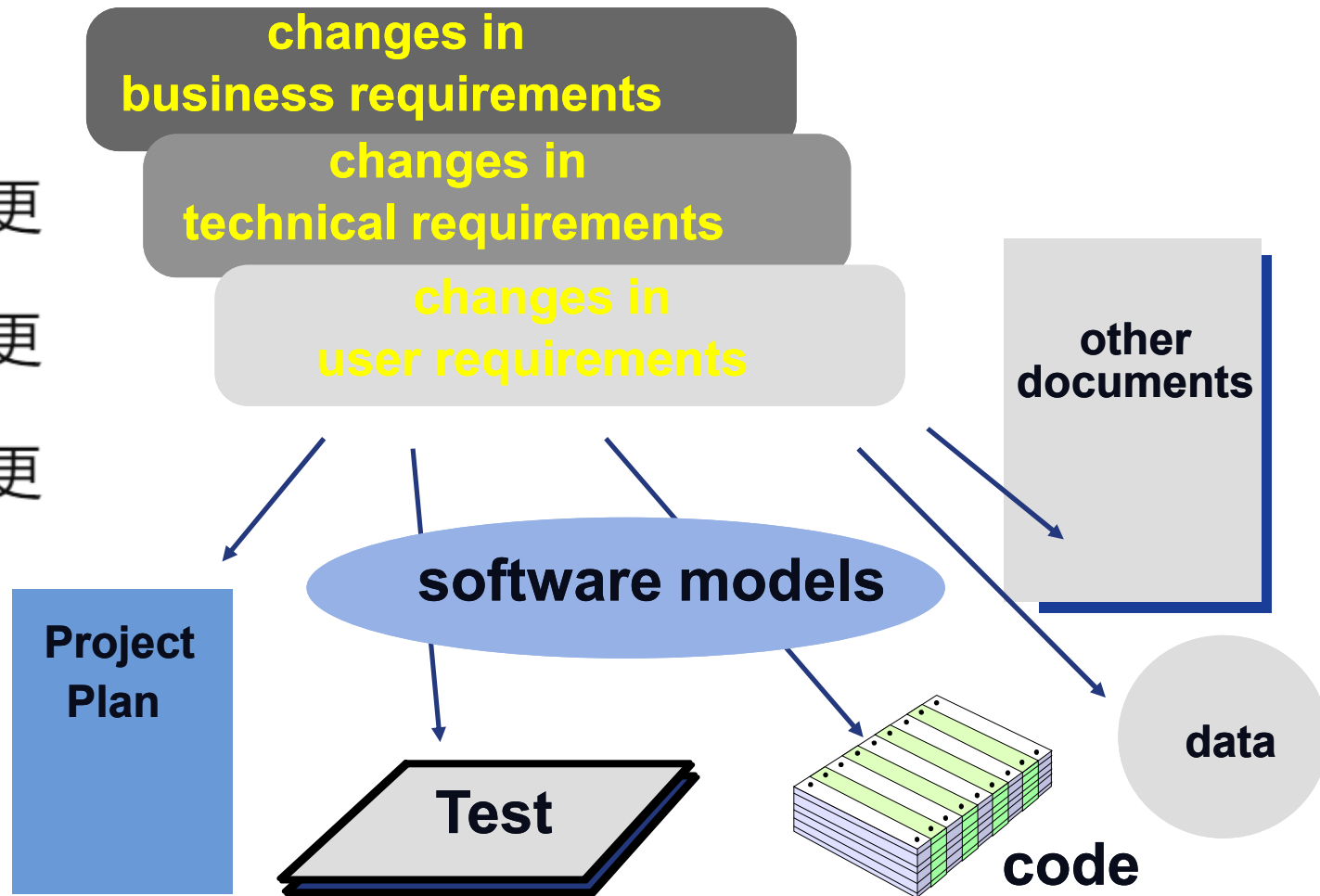
No matter where you are in the system life cycle, **the system will change, and the desire to change it will persist throughout the life cycle.**

Bersoff, et al, 1980

第一定律： 无论你处于系统生命周期的哪个阶段，系统都会发生变化，而改变它的需求会贯穿整个生命周期

What Are These Changes?

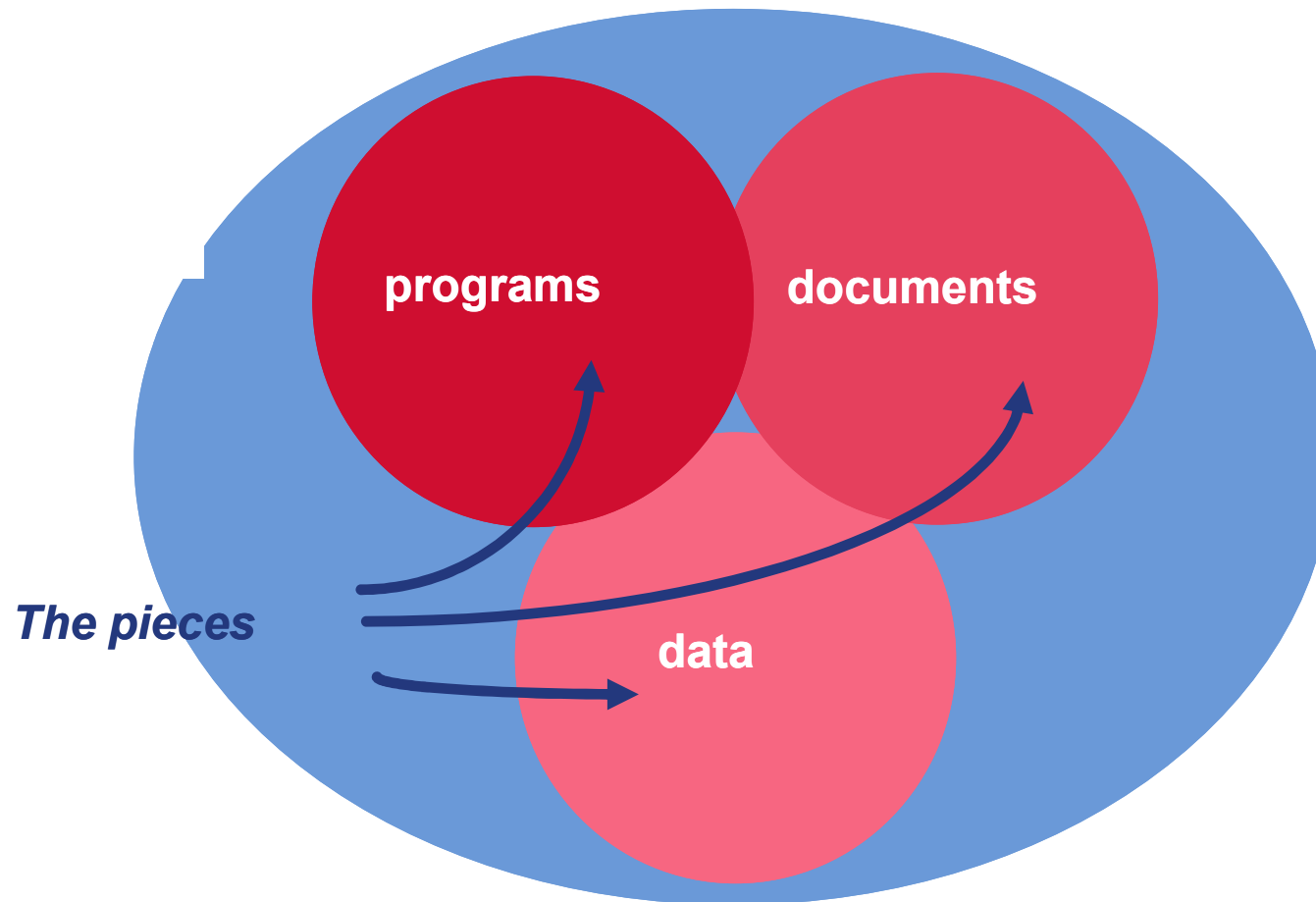
- 业务需求的变更
- 技术需求的变更
- 用户需求的变更



The Software Configuration

3. 软件配置包含三个方面

1. 程序
2. 文档
3. 数据



Baselines

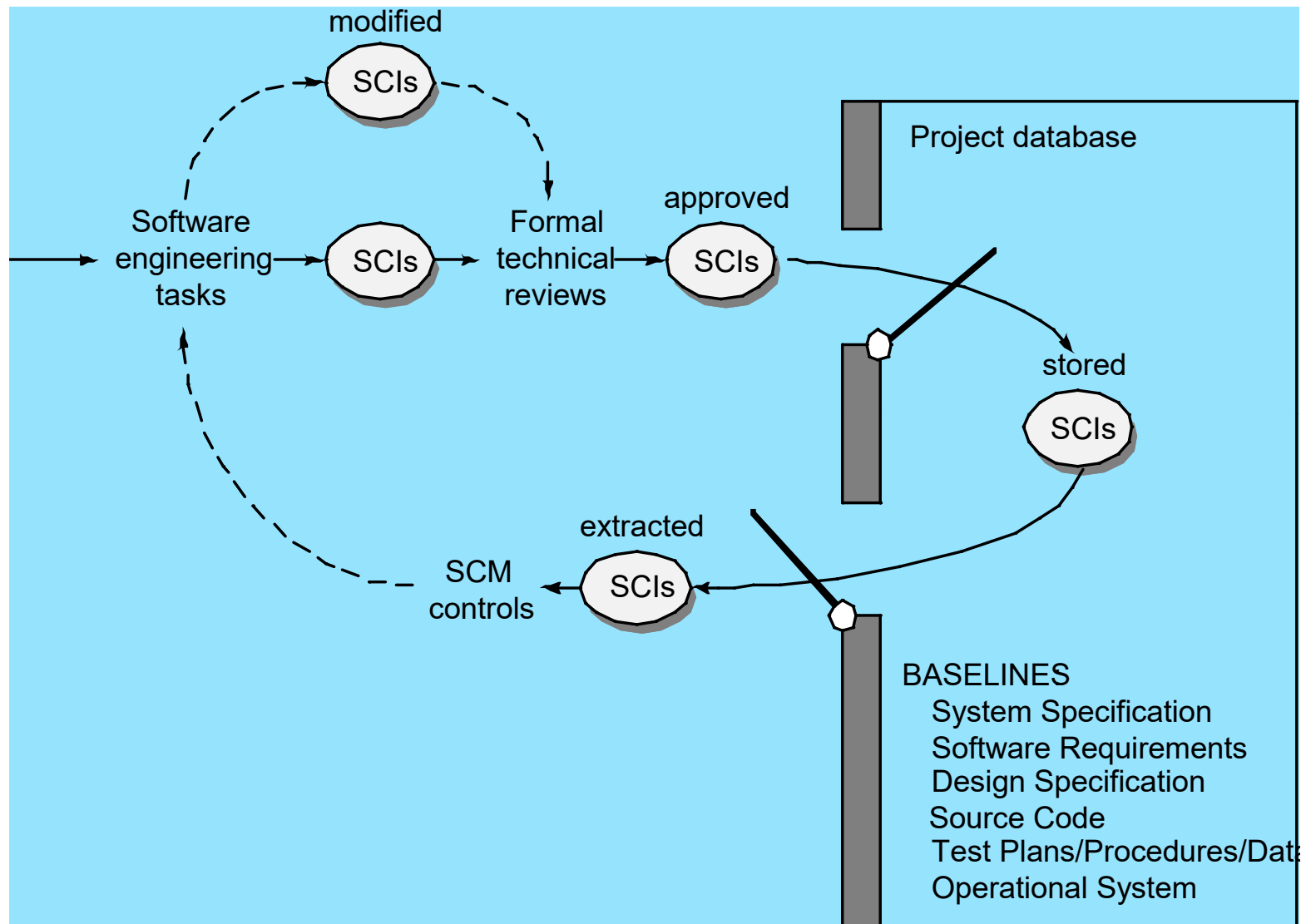
❖ **The IEEE (IEEE Std. No. 610.12-1990) defines a **baseline** as:**

- A specification or product that has been formally reviewed and agreed upon, that thereafter serves as the basis for further development, and that can be changed only through formal change control procedures.

❖ **A baseline is a milestone in the development of software that is marked by the delivery of one or more software configuration items and the approval of these SCIs that is obtained through a formal technical review**

4. 基线——已正式审查并达成一致的规范或产品，可作为进一步开发的基础，并且只能通过正式的流程才能更改
- 基线是软件开发中的一个里程碑，其标志是交付一个或多个通过正式的技术审查获得批准的软件配置项（SCI）

Baselines



SCM Repository SCM 资源库

- SCM 资源库是一组机制和数据结构，允许软件团队以有效的方式管理变更。

❖ **The SCM repository is the set of mechanisms and data structures that allow a software team to manage changes in an effective manner**

❖ **The repository performs or precipitates the following functions [For89]:** • 资源库执行或促成以下功能 [For89]:

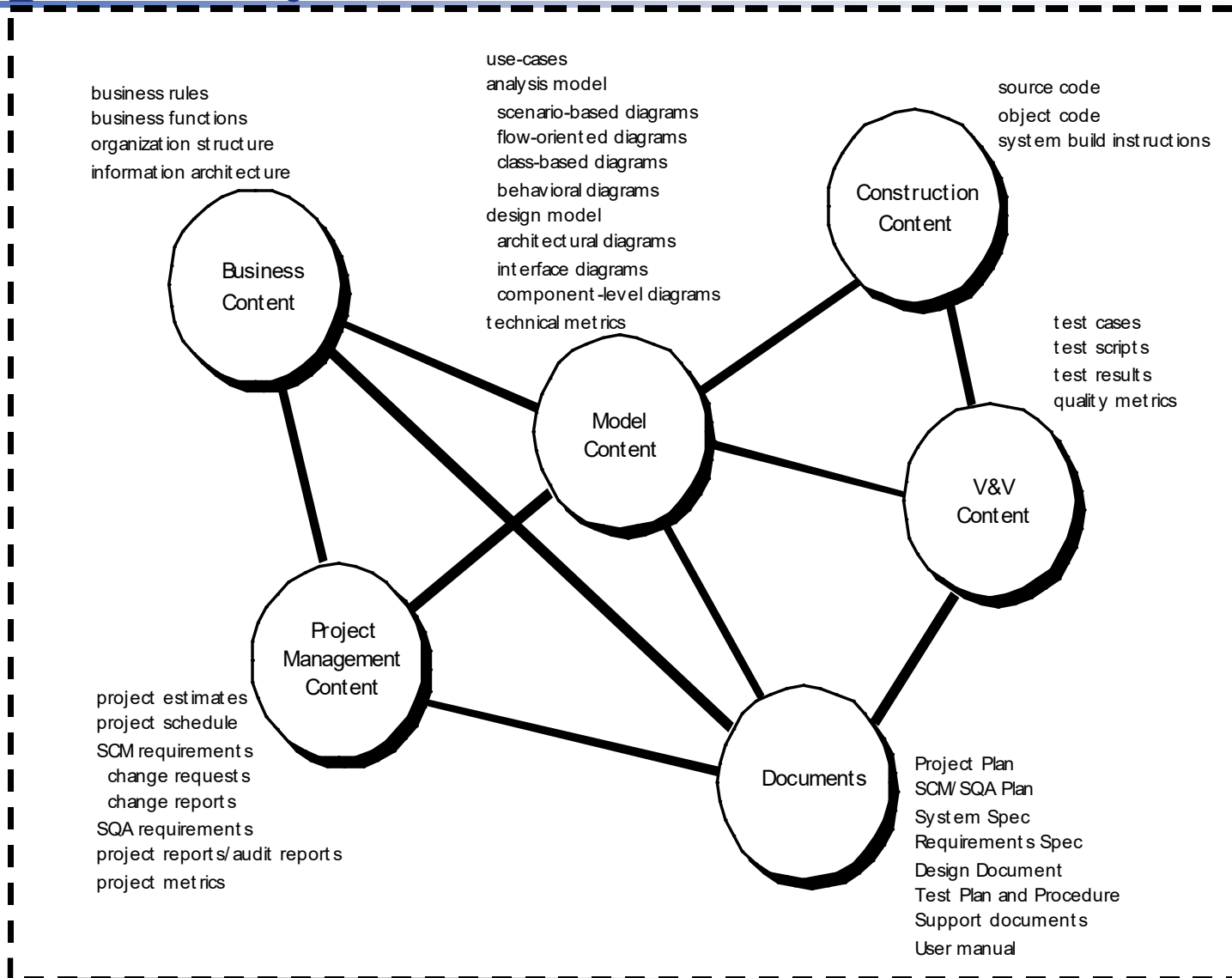
- **Data integrity**
- **Information sharing**
- **Tool integration**
- **Data integration**
- **Methodology enforcement**
- **Document standardization**

- 数据完整性
- 信息共享
- 工具集成
- 数据集成
- 方法论执行
- 文档标准化

5. SCM数据库

- 存放各种数据

Repository Content



Repository Features

❖ **Versioning.**

- saves all of these versions to enable effective management of product releases and to permit developers to go back to previous versions

❖ **Dependency tracking and change management.**

- The repository manages a wide variety of relationships among the data elements stored in it.

❖ **Requirements tracing.**

- Provides the ability to track all the design and construction components and deliverables that result from a specific requirement specification

- 版本控制——保存所有这些版本，以实现对产品发布的有效管理，并允许开发人员返回到以前的版本
- 依赖跟踪——存储库管理存储在其中的数据元素之间的各种关系
- 需求跟踪——跟踪由特定需求规范产生的所有设计和施工组件和可交付成果

Repository Features

❖ **Configuration management.**

- Keeps track of a series of configurations representing specific project milestones or production releases. Version management provides the needed versions, and link management keeps track of interdependencies.

❖ **Audit trails.**

- establishes additional information about when, why, and by whom changes are made.
- 配置管理——特定项目里程碑或生产版本的一系列配置
- 审核跟踪——建立关于何时、为什么以及由谁进行更改的信息

SCM Elements

- ❖ ***Component elements***—a set of tools coupled within a file management system (e.g., a database) that enables access to and management of each software configuration item.
- ❖ ***Process elements***—a collection of procedures and tasks that define an effective approach to change management (and related activities) for all constituencies involved in the management, engineering and use of computer software.
- ❖ ***Construction elements***—a set of tools that automate the construction of software by ensuring that the proper set of validated components (i.e., the correct version) have been assembled.
- ❖ ***Human elements***—to implement effective SCM, the software team uses a set of tools and process features (encompassing other CM elements)

- 构件元素——一组具有数据库功能的工具，使人们可以访问和管理每个软件配置项
- 过程元素——一个动作和任务集，定义变更管理的方法
- 构造元素——一组自动软件构造工具，确保正确装配所有构件
- 人员元素——为了实施有效的 SCM，软件团队使用一组工具和流程特征（包括其他 CM 元素）

The SCM Process

Addresses the following questions ...

- ❖ **How does a software team identify the discrete elements of a software configuration?**
- ❖ **How does an organization manage the many existing versions of a program (and its documentation) in a manner that will enable change to be accommodated efficiently?**
- ❖ **How does an organization control changes before and after software is released to a customer?**
- ❖ **Who has responsibility for approving and ranking changes?**
- ❖ **How can we ensure that changes have been made properly?**
- ❖ **What mechanism is used to appraise others of changes that are made?**

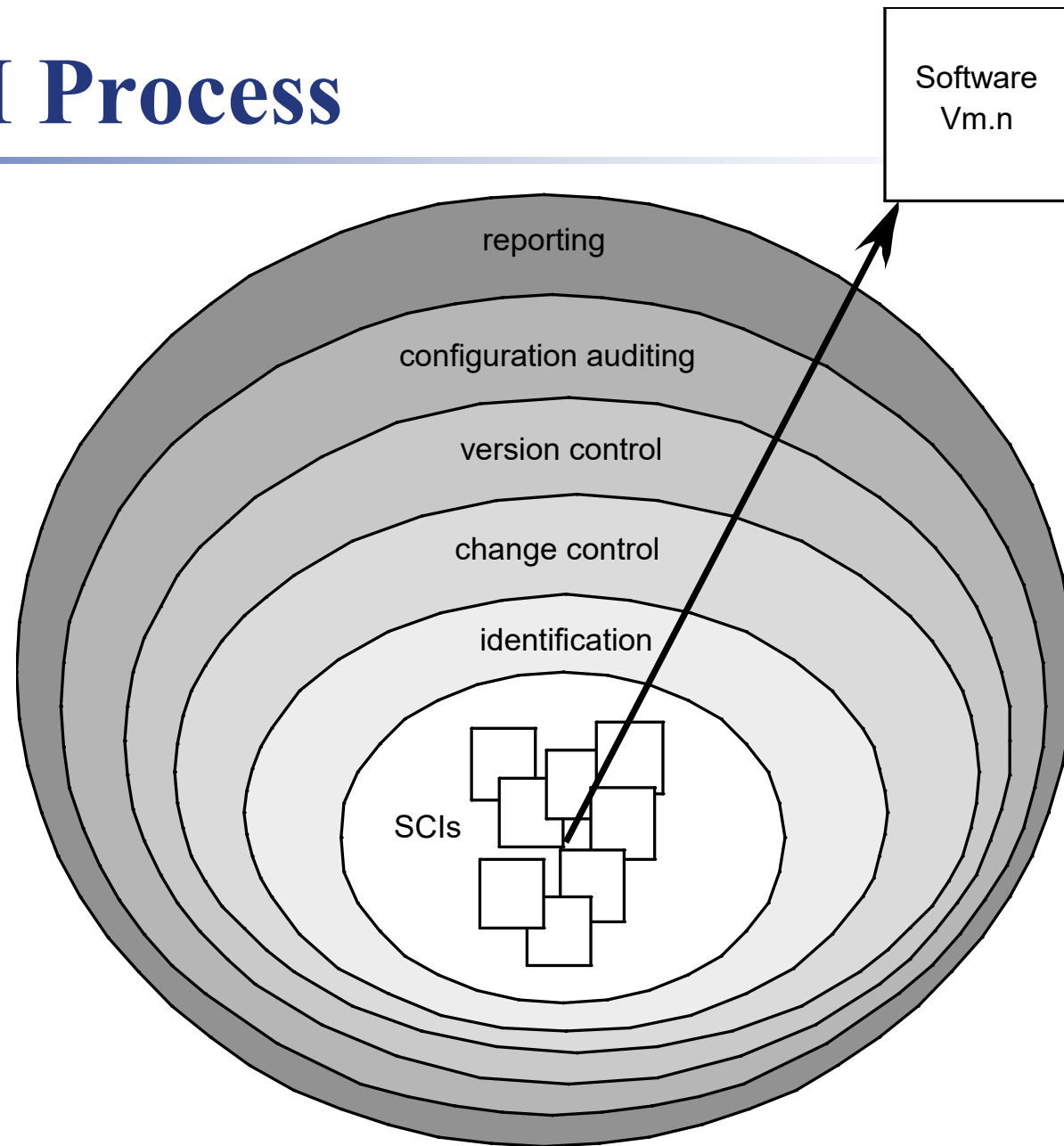
SCM过程

8. SCM过程 (CCA:变更控制授权人 ECO: 工程变更工单)

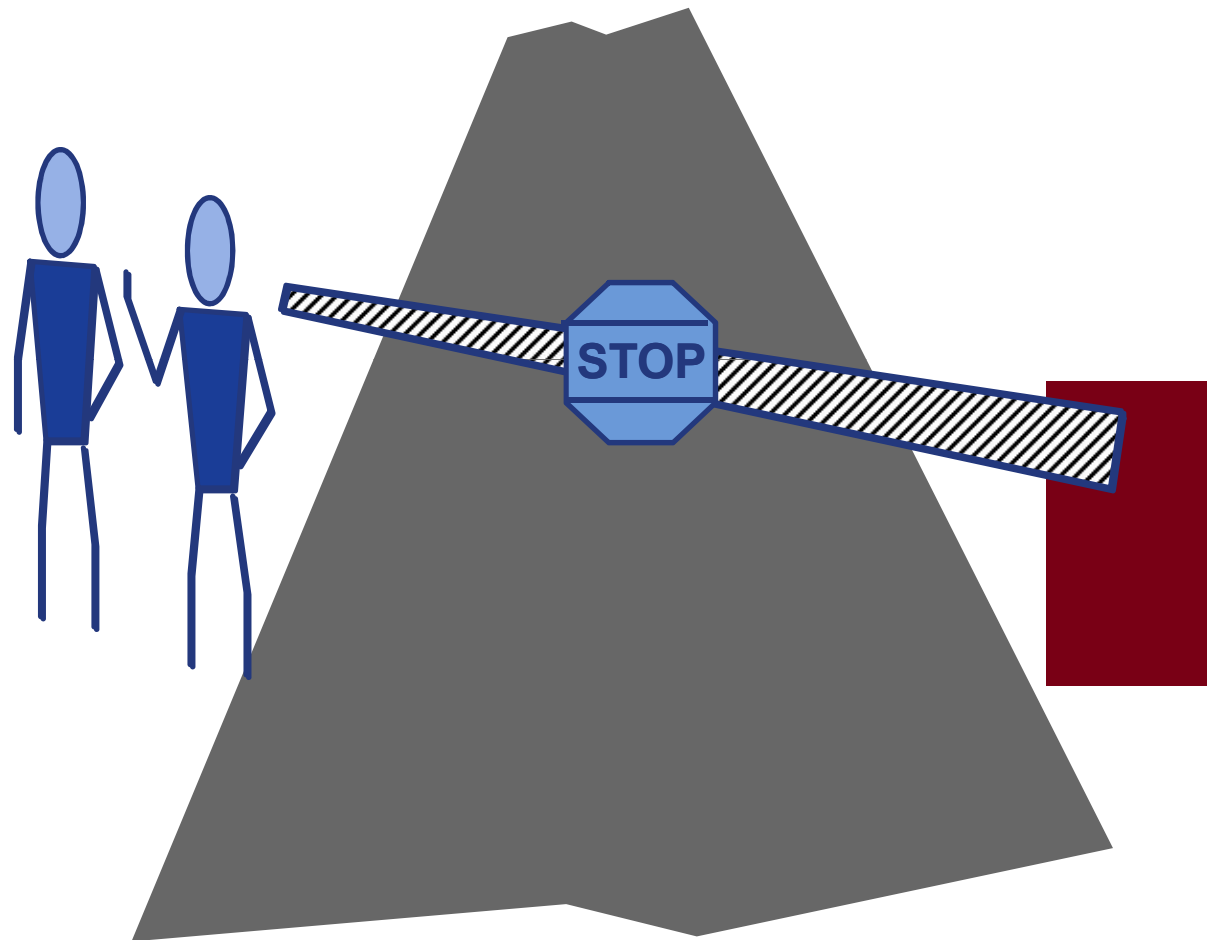
- 5个SCM任务

- 标识
- 版本控制——结合程序和工具来管理在软件过程中创建的不同版本的配置对象
 - 版本控制系统实现或直接集成了四个主要功能
 1. 存储所有相关配置对象的项目数据库（存储库）
 2. 存储配置对象的所有版本
 3. 使软件工程师能够收集所有相关配置对象并构建软件的特定版本的制作工具
 4. 问题跟踪（也称为错误跟踪）功能，使团队能够记录和跟踪与每个配置对象相关的所有未决问题的状态
- 变更控制
- 配置审核
- 报告

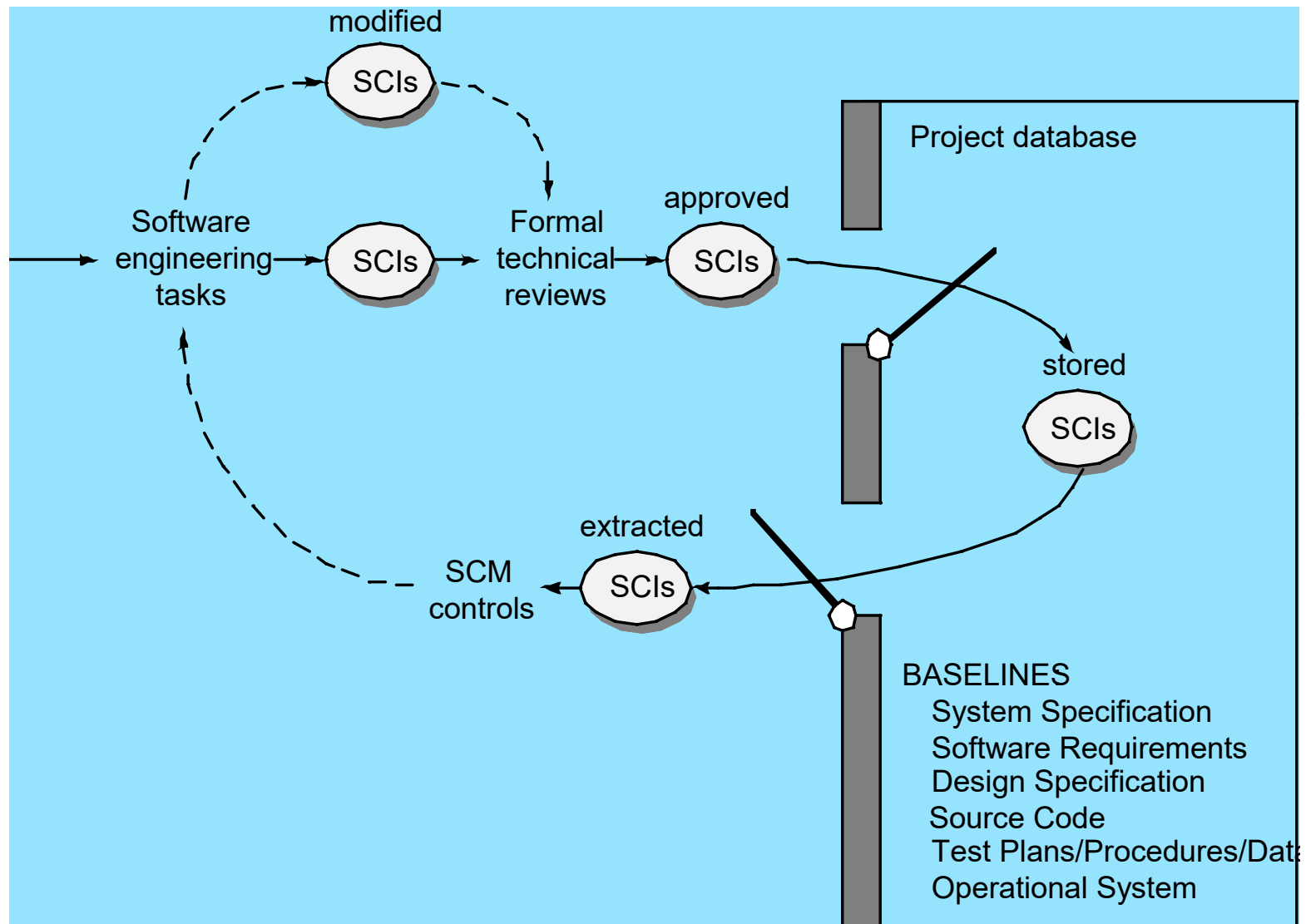
The SCM Process



Change Control

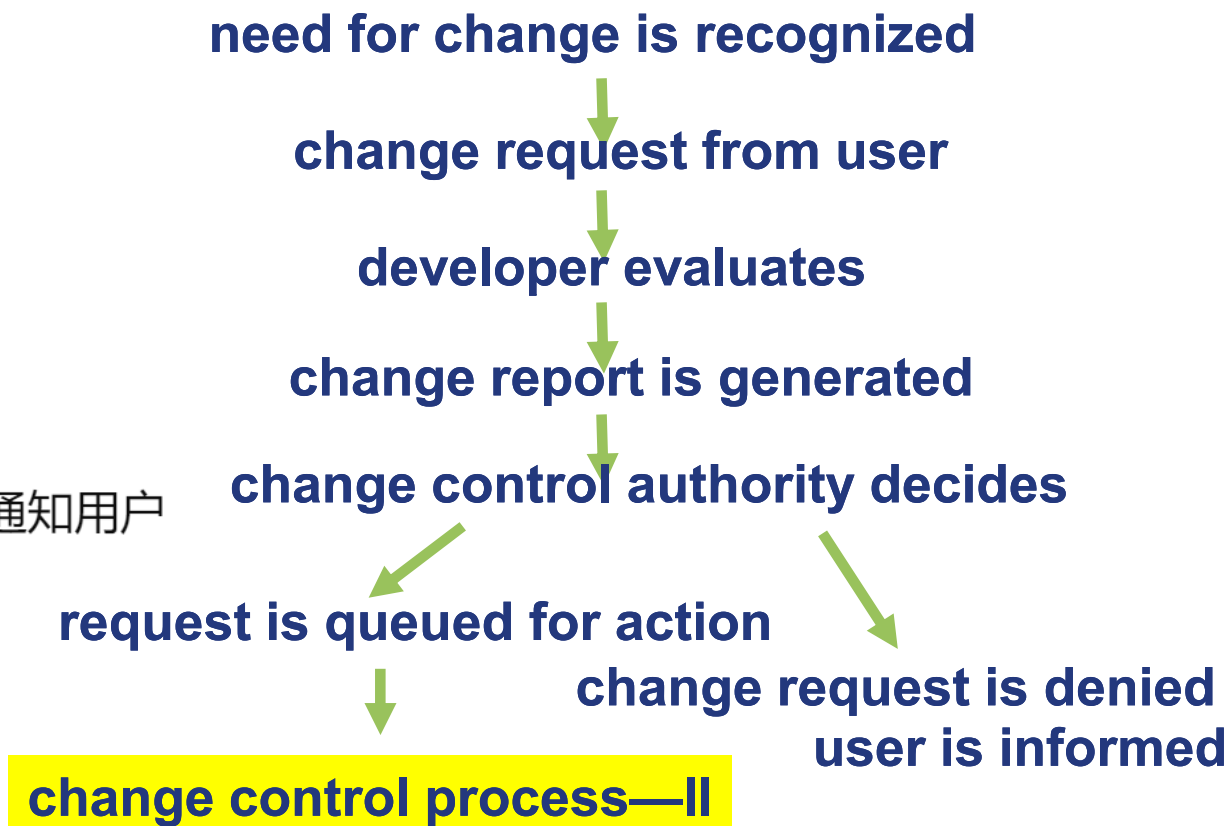


Change Control

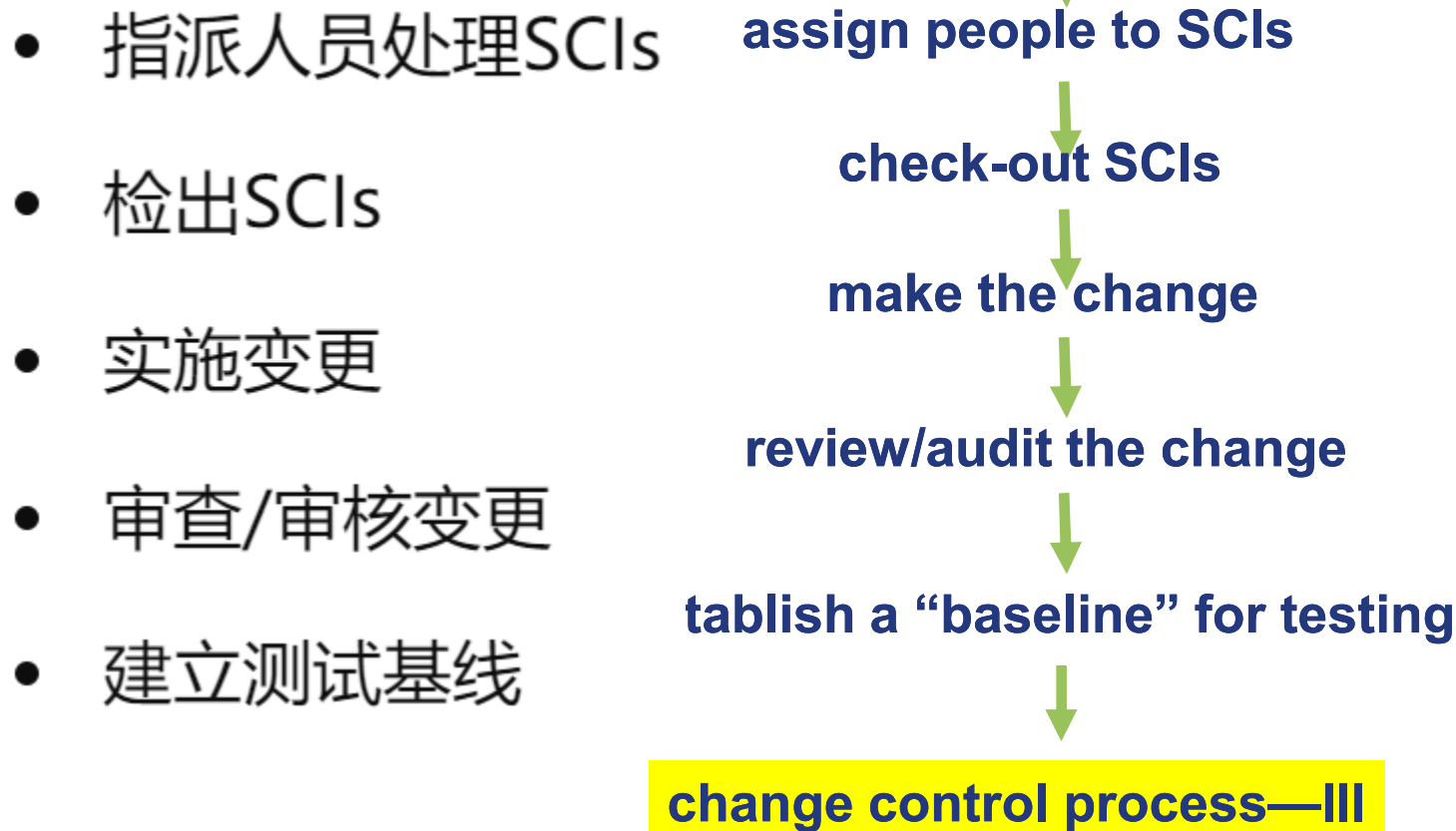


Change Control Process—I

- 变更需求被识别
- 用户提交变更请求
- 开发人员进行评估
- 生成变更报告
- 变更控制机构决定
 - 请求排队等待处理
 - 变更请求被拒绝并通知用户



Change Control Process-II



Change Control Process-III



Version Control

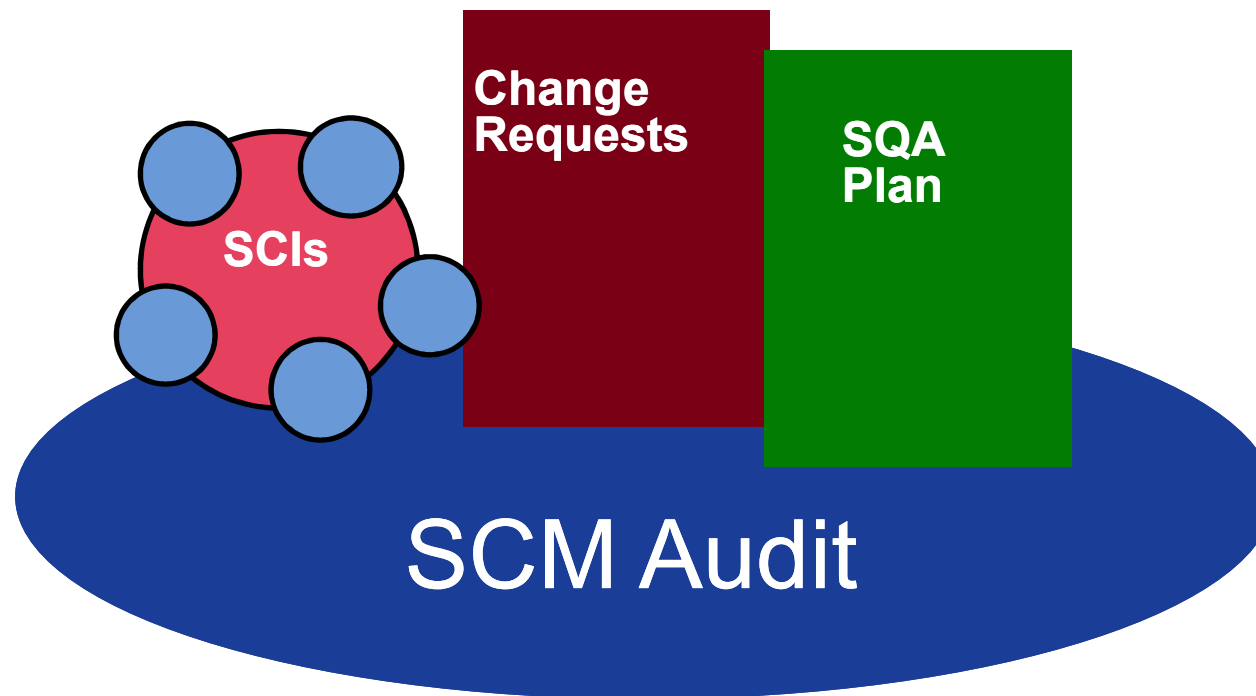
- ❖ **Version control combines procedures and tools to manage different versions of configuration objects that are created during the software process**
- ❖ **A version control system implements or is directly integrated with four major capabilities:**
 - **a *project database (repository)* that stores all relevant configuration objects**
 - **a *version management* capability that stores all versions of a configuration object (or enables any version to be constructed using differences from past versions);**
 - **a *make facility* that enables the software engineer to collect all relevant configuration objects and construct a specific version of the software.**
 - **an *issues tracking* (also called *bug tracking*) capability that enables the team to record and track the status of all outstanding issues associated with each configuration object.**

Version Control

版本控制

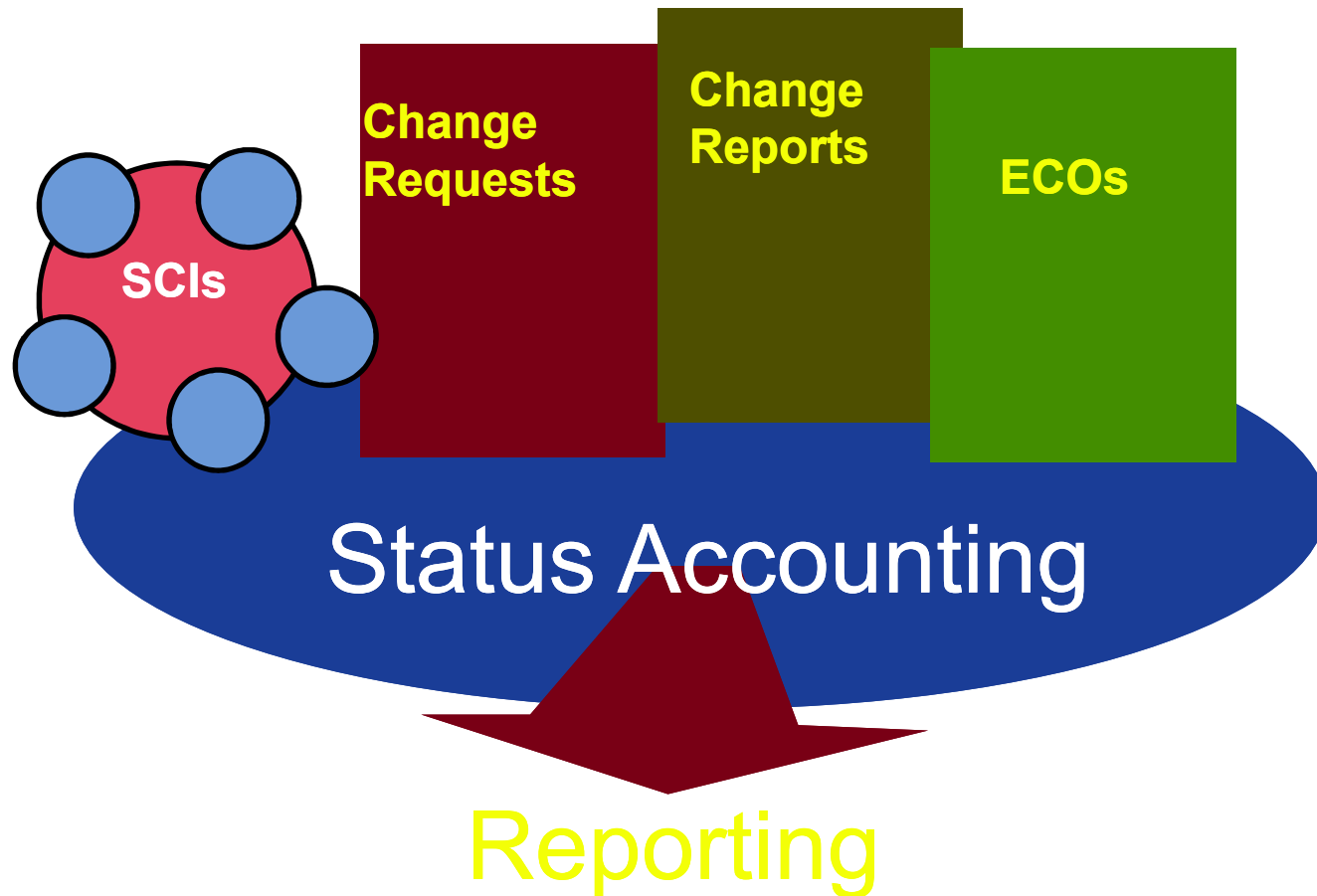
- 版本控制结合程序和工具，以在软件过程中管理不同版本的配置对象
- 版本控制系统实现或直接集成了四个主要功能：
 - 存储所有相关配置对象的项目数据库（资源库）
 - 存储配置对象所有版本的版本管理功能（或使用过去版本的差异构建任何版本）
 - 使软件工程师能够收集所有相关配置对象并构建软件特定版本的生成功能
 - 记录和跟踪与每个配置对象相关的所有未解决问题状态的问题跟踪功能（也称为错误跟踪）

Auditing 配置审核



Status Accounting

报告





Southeast University



Thank You !

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